

CSE 112 – Lab #6 Object Oriented Exception Handling with Classes

Ref: page 1028 – Rectangle and Program 16-2, and 16-5

Modify the Circle Class from the earlier lab to add exception handling for negative radius inputs.

Modify the main program to use a try/catch as shown below. This will require creating a **NegativeRadius class** in the Circle class. The NegativeRadius class will have a constructor that receives the radius and stores it in a **member variable (value)** (Ref 16.1) and has a member **function *getValue()*** that returns the value. This chronology requires creating an instance of the circle before trying to set the radius in the try block of main (see below). **Add the loop as shown below** so that multiple iterations of the program can be run. Submit a text file with the modified main, and header file.

```
int main()
{
    double radius = 0; // To hold a radius
    char tryAgain = 'y';

    Circle userCircle;

    while(tryAgain == 'y' || tryAgain == 'Y')
    {
        cout << "Enter the circle's radius: ";
        cin >> radius;

        try
        {
            userCircle.setRadius(radius);
            cout << setprecision(2) << fixed;
            cout << "Radius: " << userCircle.getRadius() << endl;
            cout << "Area : " << userCircle.getArea() << endl;
            cout << "Diameter: " << userCircle.getDiameter() << endl;
            cout << "Circumference: " << userCircle.getCircumference() << endl << endl;
        }
        catch (Circle::NegativeRadius e)
        {
            cout << "Error: " << e.getValue() << "is not a valid radius." << endl << endl;
        }

        cout << "Enter Y to run again:";
        cin >> tryAgain;
    }

    return 0;
}
```

Grading will be based on meeting all of the lab requirements, adherence to programming standards, operation and accurate output, and programming style including white space and indentation.